

Question # 1

 Revisit

Choose the best option

Suppose the distribution of salaries in a company X has median \$35,000, and 25th and 75th percentiles are \$21,000 and \$53,000 respectively. Would a person with Salary \$1 be considered an Outlier?

- ☐ Yes
- ☐ No
- ☒ More information is required
- ☐ none of these choices

Question # 2

 Revisit

In performing a hypothesis test where the null hypothesis is that the population mean is 45 against the alternative hypothesis that the population mean is not equal to 45, a random sample of 17 items is selected. The sample mean is 47.6 and the sample standard deviation is 3.3. It can be assumed that the population is normally distributed. The degrees of freedom associated with this are:

Choose the best option

- ☐ 2
- ☐ 15
- ☐ 16
- ☐ 17

Question # 5

 Revisit

In performing a hypothesis test where the null hypothesis is that the population mean is 6.9 against the alternative hypothesis that the population mean is not equal to 6.9, a random sample of 16 items is selected. The sample mean is 7.1 and the sample standard deviation is 2.4. It can be assumed that the population is normally distributed. The observed "t" value for this problem is:

Choose the best option

- ☐ 0.05
- ☐ 0.2
- ☐ 0.33
- ☒ 1.33

Question # 4

A simulation uses the logical relationships and mathematical expressions of the

 Revisit

Choose the best option

- ☐ Real system
- ☐ Computer model
- ☐ Performance measures
- ☐ Inferences

Question # 3

 Revisit

In simple linear regression, which of the following statements indicates there is no linear relationship between the variables x and y ?

Choose the best option

- ☐ Coefficient of determination is -1.0 .
- ☐ Coefficient of correlation is 0.0 .
- ☐ Sum of squares for error is 0.0 .
- ☐ none of these choices



Question # 6

In testing hypotheses, the researcher initially assumes:

 Revisit

Choose the best option

- ☐ the alternative hypothesis is true.
- ☐ the null hypothesis is true
- ☐ errors cannot be made
- ☐ the population parameter of interest is known



Question # 7

Revisit

Whenever hypotheses are established such that the alternative hypothesis is ">", then this would be a _____.

Choose the best option

- ☐ two-tailed test
- ☐ Type II test
- ☐ one-tailed test
- ☐ Type I test

Aufgabe 1

(10 Punkte)

(10 Punkte)

(10 Punkte)

- ☐ In einem Ring R gilt: $a^2 = a$ für alle $a \in R$. Zeigen Sie, dass R ein Körper ist.
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- ☒ In einem Ring R gilt: $a^2 = a$ für alle $a \in R$. Zeigen Sie, dass R ein Körper ist.

(10 Punkte)

Question # 8

🔄 Revisit

The previous probabilities in Bay's Theorem that are changed with the help of new available information are classified as

Choose the best option

- ☐ Independent probabilities
- ☐ Posterior probabilities
- ☐ Interior probabilities
- ☐ Dependent probabilities

Question # 12

Revisit

For a given level of significance, if the sample size is increased, the power of the test will

Choose the best option

- ☐ increase
- ☐ stay the same
- ☐ decrease
- ☒ none

Clear Response

● 2013 年 10 月 1 日起实施

For the first 2000 years of civilization, the development of science, history, and philosophy was slow and steady. It was not until the 17th century that the scientific revolution began to take hold, and the world began to change rapidly. The 18th century saw the rise of the Enlightenment, and the 19th century saw the Industrial Revolution. The 20th century was a time of great change, with the rise of the atomic age, the space age, and the computer age. The 21st century is a time of great uncertainty, with the rise of the digital age, the climate change crisis, and the global pandemic.

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- ☐ Sociolinguistics
☐ Phonetics
☐ Morphology
☒ **Acquisition of Second Language**
☐ Syntax- Semantics

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When all the following conditions are met, you can be downloading a new language pack for Windows 7 without any issues:

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• **Stressoren** – Stress verursachende Ereignisse

- ☒ ১. অসম্পূর্ণ

Question # 15

Revisit

Choose the best option

When a true null hypothesis is rejected, the researcher has committed a:

- ☐ Type II error
- ☒ Type I error
- ☐ rejection error
- ☐ error in judgment
- Clear Response

Question # 14

🔄 Revisit

A statistics professor asked students in a class their ages. On the basis of this information, the professor states that the average age of all the students in the university is 21 years. This is an example of

Choose the best option

- ☒ census
- ☐ descriptive statistics
- ☐ an experiment
- ☐ statistical inference

Question # 13

Revisit

The standard deviation of a sample of 100 observations equals 64. The variance of the sample equals

Choose the best option

- ☐ 8
- ☐ 10
- ☐ 6400
- ☒ 4096

Clear Response

Question # 18

🔄 Revisit

Choose the best option

The expected value of Y is a linear function of the $X(X_1, X_2, \dots, X_n)$ variables and regression line is defined as: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$. Which of the following statement(s) are true? 1) If X_i changes by an amount ΔX_i , holding other variables constant, then the expected value of Y changes by a proportional amount $\beta_i \Delta X_i$, for some constant β_i (which in general could be a positive or negative number). 2) The value of β_i is always the same, regardless of values of the other X's. 3) The total effect of the X's on the expected value of Y is the sum of their separate effects. Note: Features are independent of each others (zero interaction).

- ☐ 1 and 2
☐ 1 and 3
☐ 2 and 3
☐ 1, 2 and 3

Question # 17

🔄 Revisit

In simple linear regression, the coefficient of correlation r and the least squares estimate b_1 of the population slope β_1 :

Choose the best option

- ☐ must be equal.
- ☐ must have the same sign.
- ☐ are not related.
- ☒ none of these choices

Clear Response

Question # 16

🔄 Revisit

Choose the best option

Generally, which of the following method(s) is used for predicting continuous dependent variable?

1. Linear Regression
2. Logistic Regression

- ☐ 1 and 2
- ☐ only 1
- ☐ only 2
- ☒ none of these choices
- Clear Response

Question # 27

 Revisit

Choose the best option

Which of the following indicates a fairly strong relationship between X and Y?

- ☐ Correlation coefficient = 0.9
- ☐ The p-value for the null hypothesis Beta coefficient = 0 is 0.0001
- ☐ The t-statistic for the null hypothesis Beta coefficient = 0 is 30
- ☒ none of these choices

Clear Response

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18 19 20 21 22 23 24 25 26 27 28 29 30 < 26 of 40 >

Total 00:57:01
Section 00:57:01 Finish Test

Question # 26

Which of the following criteria cannot be used to determine the number of factors in an EFA?

Revisit

Choose the best option

- ☐ Asking a group of researchers before the analysis
- ☐ Eigenvalue rule
- ☐ Scree test
- ☒ Parallel analysis

Clear Response

Question # 19

🔄 Revisit

The strength of the linear relationship between two numerical variables may be measured by the

Choose the best option

- ☐ scatter diagram
- ☐ coefficient of correlation
- ☐ slope
- ☐ Y intercept

Question # 31

Which of the following is Step 1 for Performing Simulation Analysis

 Revisit

Choose the best option

- ☐ choose input variables
- ☐ create entities for the simulation process
- ☐ prepare a problem statement
- ☐ determine the output variables

23 24 25 26 27 28 29 30 31 32 33 34 35 < 31 of 40 >

Total 00:58:45
Section 00:58:45 Finish Test

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18 19 20 21 22 23 24 25 26 27 28 29 30

Total 00:58:51
Section 00:58:51 Finish Test

< 29 of 40 >

All 28 12

Revisit

Choose the best option

- ☐ Type I test
- ☐ two-tailed test
- ☐ one-tailed test
- ☐ Type II test

Question # 29

Whenever hypotheses are established such that the alternative hypothesis is "not equal to", then the test is a:

Question # 28

What will a factor loading in an orthogonal solution represent?

Revisit

Choose the best option

- ☐ Correlation
- ☐ Partial correlation
- ☐ Multiple correlation
- ☒ Eigenvalue
- [Clear Response](#)

Question # 33

Revisit

The statistical distribution used for testing the difference between two population variances is the ____ distribution.

Choose the best option

- ☐ t
- ☐ standardized normal
- ☐ binomial
- ☒ F

Clear Response

Question # 32

Revisit

Problems which seek to maximise or, minimise profit or, cost form a general class of problems called

Choose the best option

- ☐ Simple problems
- ☐ Difficult problems
- ☐ Optimisation problems
- ☐ Non-linear problems